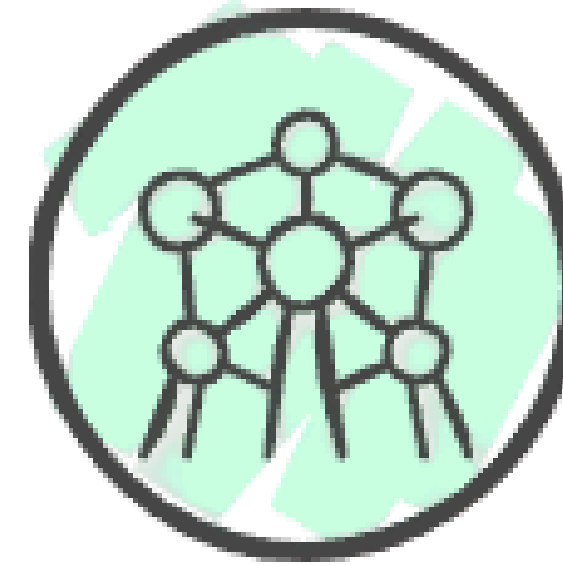


Choose the best RAG architecture for multi-hop question answering.



Naive RAG

Vector search for broad
retrieval

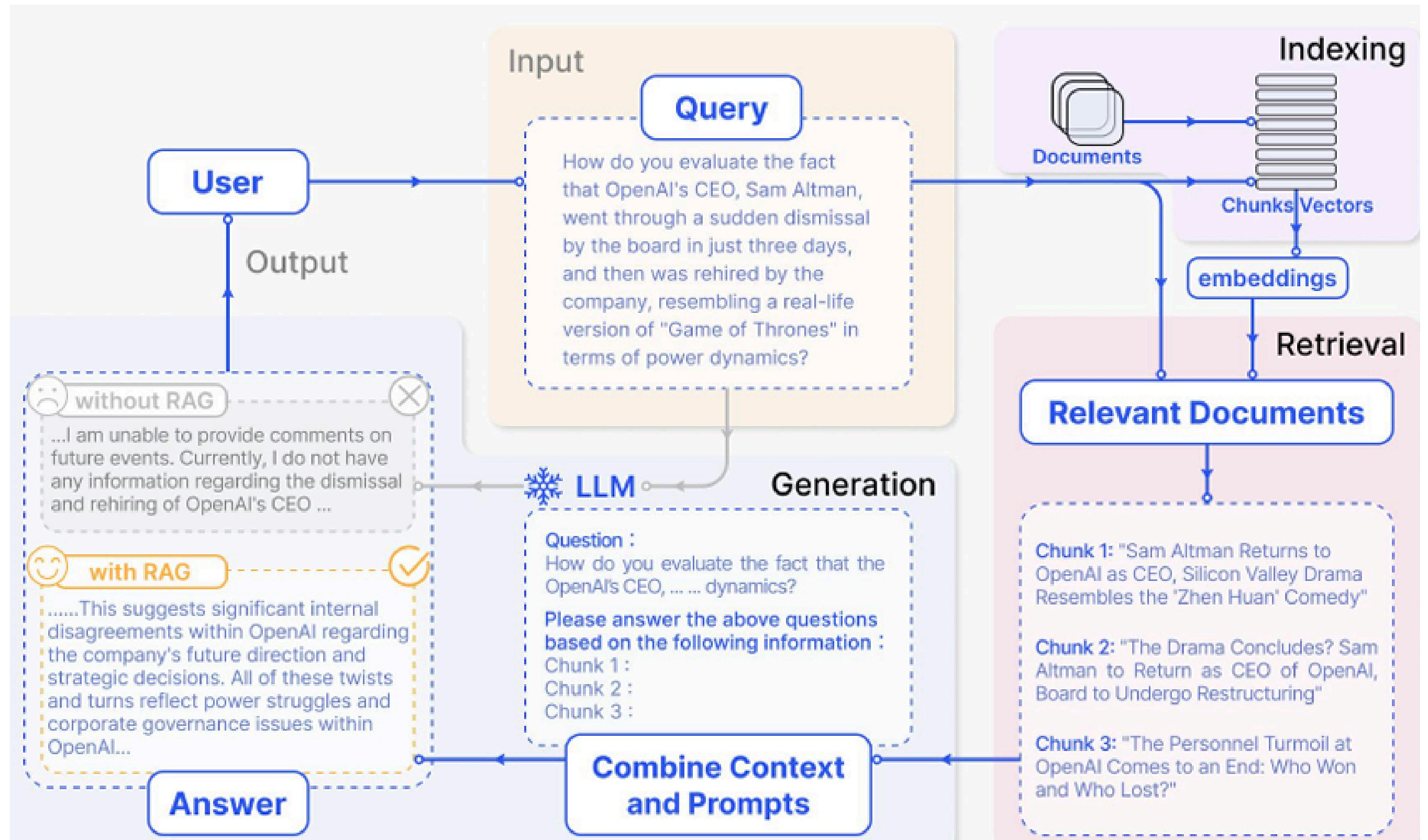


KG-RAG

Knowledge graph for structured
relationships

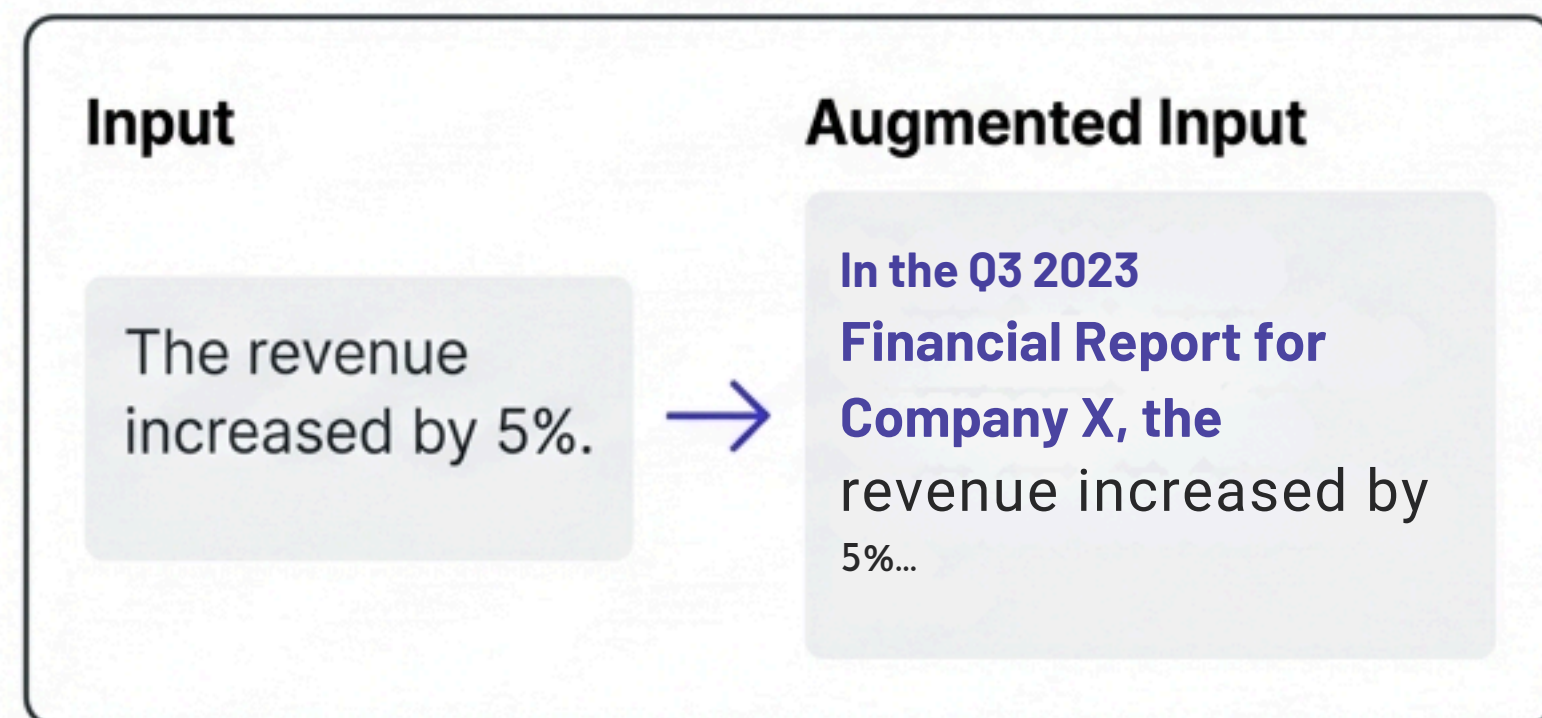
Retrieval Augmented Generation

Typical RAG



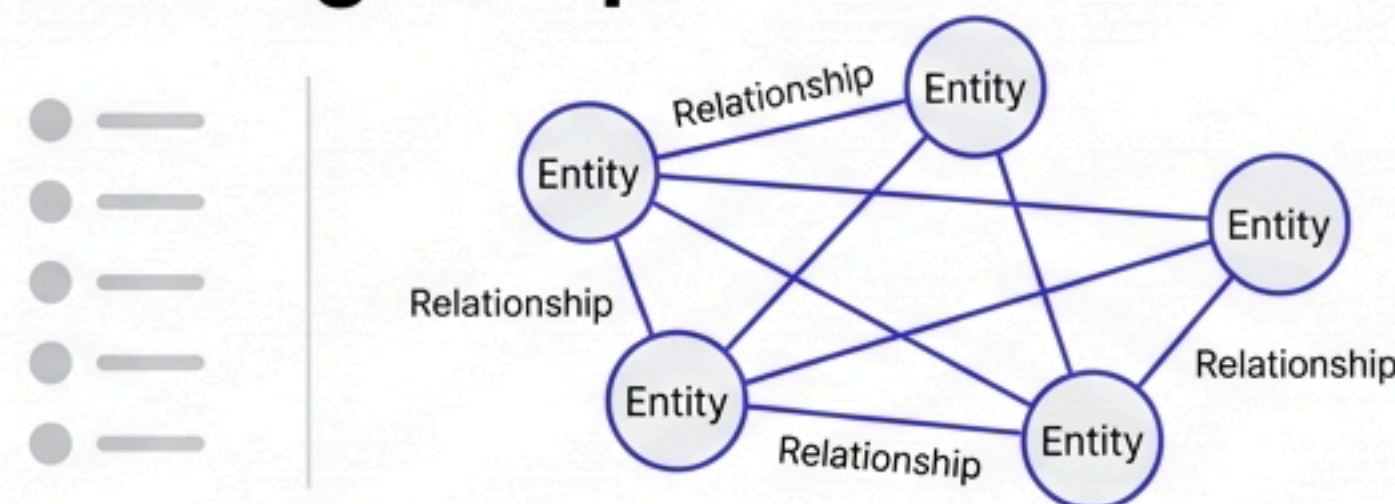
Vector retrieval – KG retrieval

Contextual Retrieval



LLM prepends context to every chunk during indexing.

Knowledge Graphs



Maps entity relationships. Best for highly interconnected data.

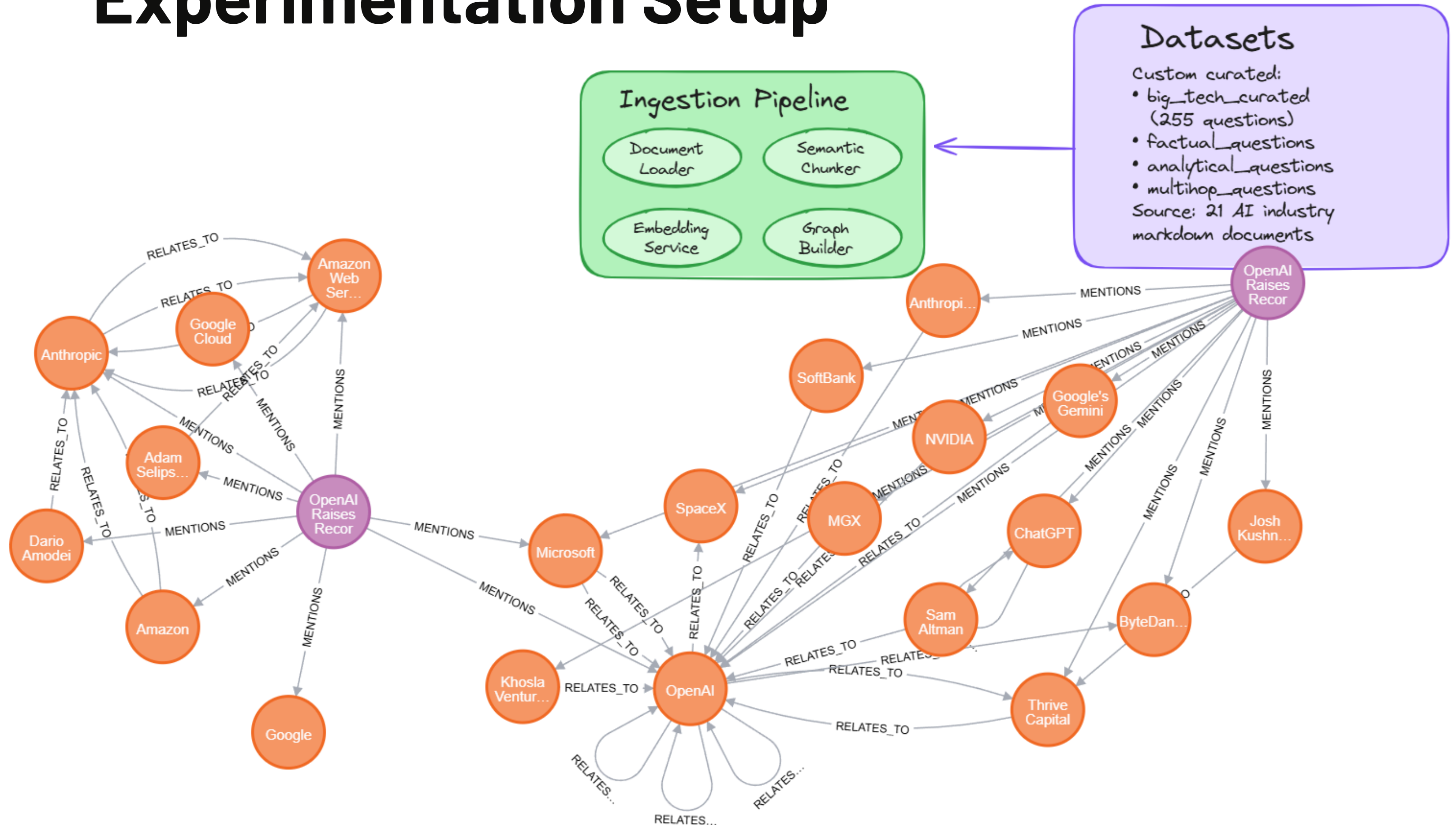
```
import graffiti
```

Trade-offs

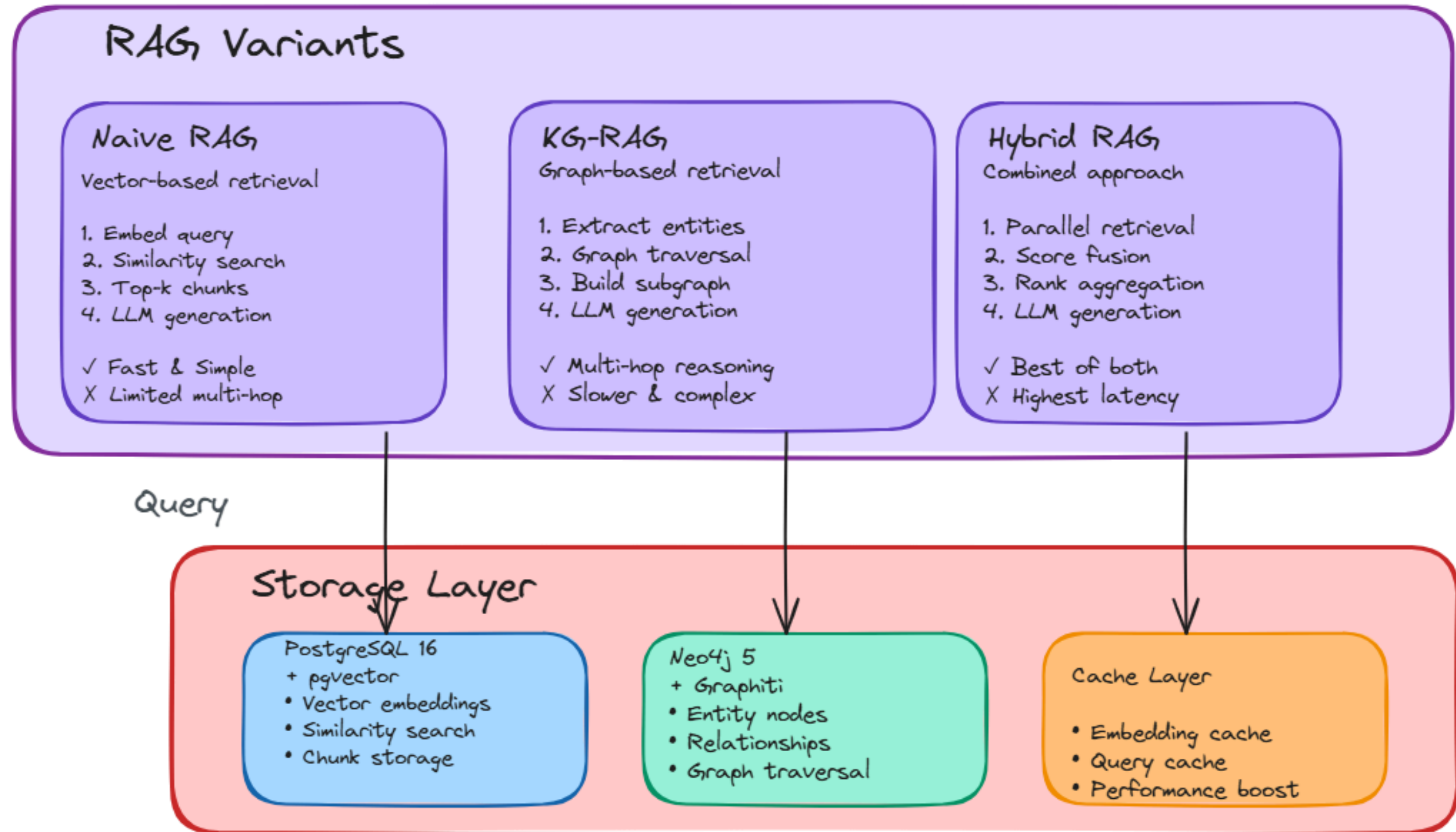
Latency (Prep): Very High

Retrieval Accuracy: Very High

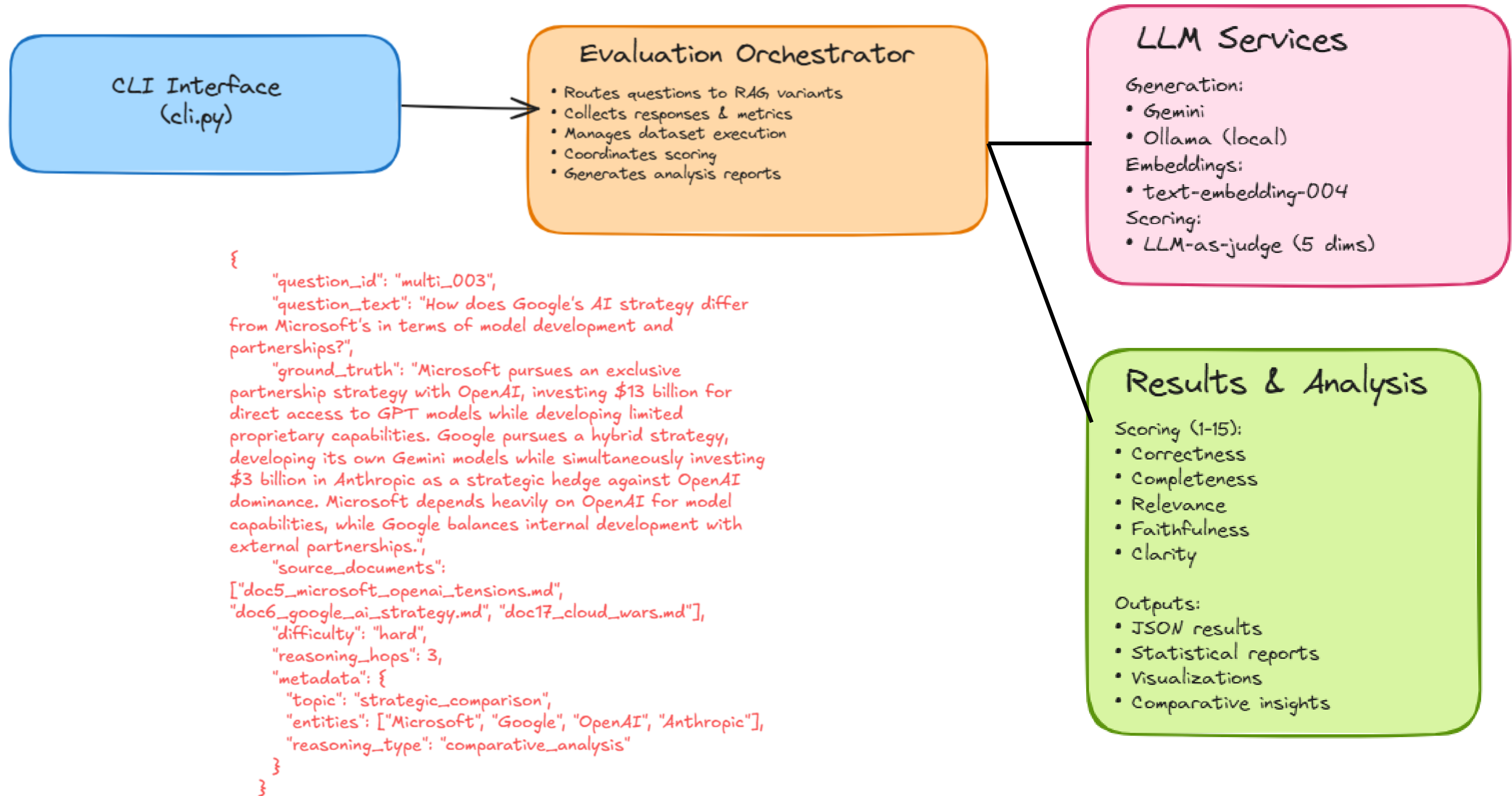
Experimentation Setup



Experimentation Setup



Experimentation Setup



Agent retrieval

```
17:52:59 - INFO - HybridRAG initialized
Initializing LangGraph agent...

› You: how does microsoft relate to openAI

🤖 Agent: Processing query...

🔍 Analyzing query...
💡 Decision: RETRIEVE from knowledge base
📖 Retrieving from knowledge base...
17:53:11 - INFO - Vector search with similarity_threshold=0.6, top_k=5
17:53:11 - INFO - Vector search returned 4 rows
17:53:11 - INFO - Retrieved 4 chunks (similarity >= 0.6, capped at 20)
17:53:12 - INFO - Entity search returned 10 edges/facts from Graphiti (hybrid BM25 + vector search with RRF reranking)
17:53:12 - INFO - Extracted 7 unique entities from 10 edges
17:53:12 - INFO - Expanded to 22 entities total (+15 via multi-hop, weights: direct=7, 1-hop=15, 2-hop=0)
17:53:12 - INFO - KG retrieval: 18 chunks, avg direct_recall=0.48, avg weighted_count=4.7, avg direct_count=3.3, entities=22, min_ratio=0.3
17:53:12 - INFO - Retrieved 18 chunks via KG (recall >= 0.3, capped at 20)
17:53:12 - INFO - Vector: 4 chunks, KG: 18 chunks
17:53:12 - INFO - Hybrid retrieval returned 20 chunks (vector_only=2, kg_only=16, both=2) | Score contribution: vector=19.4%, kg=80.6%
✓ Retrieved 20 chunks
🌀 Generating answer...
✅ Answer generated

=====
✓ Answer:
=====

Microsoft has a $13 billion strategic partnership with OpenAI, but also lists OpenAI as a competitor in AI and search markets. Microsoft in
vested $1 billion in OpenAI in 2019, $10 billion in 2023 for a 49% stake, and an additional $3 billion in 2024. Despite the investment, eit
her party can exit the partnership under specific conditions. Microsoft's Azure growth is partially driven by OpenAI integration, and Bing'
s market share increased since ChatGPT integration. 65% of OpenAI's API usage runs on Azure infrastructure, and Microsoft provides compute
subsidies. Microsoft offered to hire the entire OpenAI team when Sam Altman was fired.

Agent Path: analyze → retrieve → answer
Iterations: 3, Time: 4314ms

› You: 
```

Quantitative Results

RAG EVALUATION REPORT

Dataset: big_tech_curated

Evaluation ID: 20260112_191743_b0d908ff

Timestamp: 2026-01-12T19:17:43.929623

Number of Questions: 255

METHOD COMPARISON

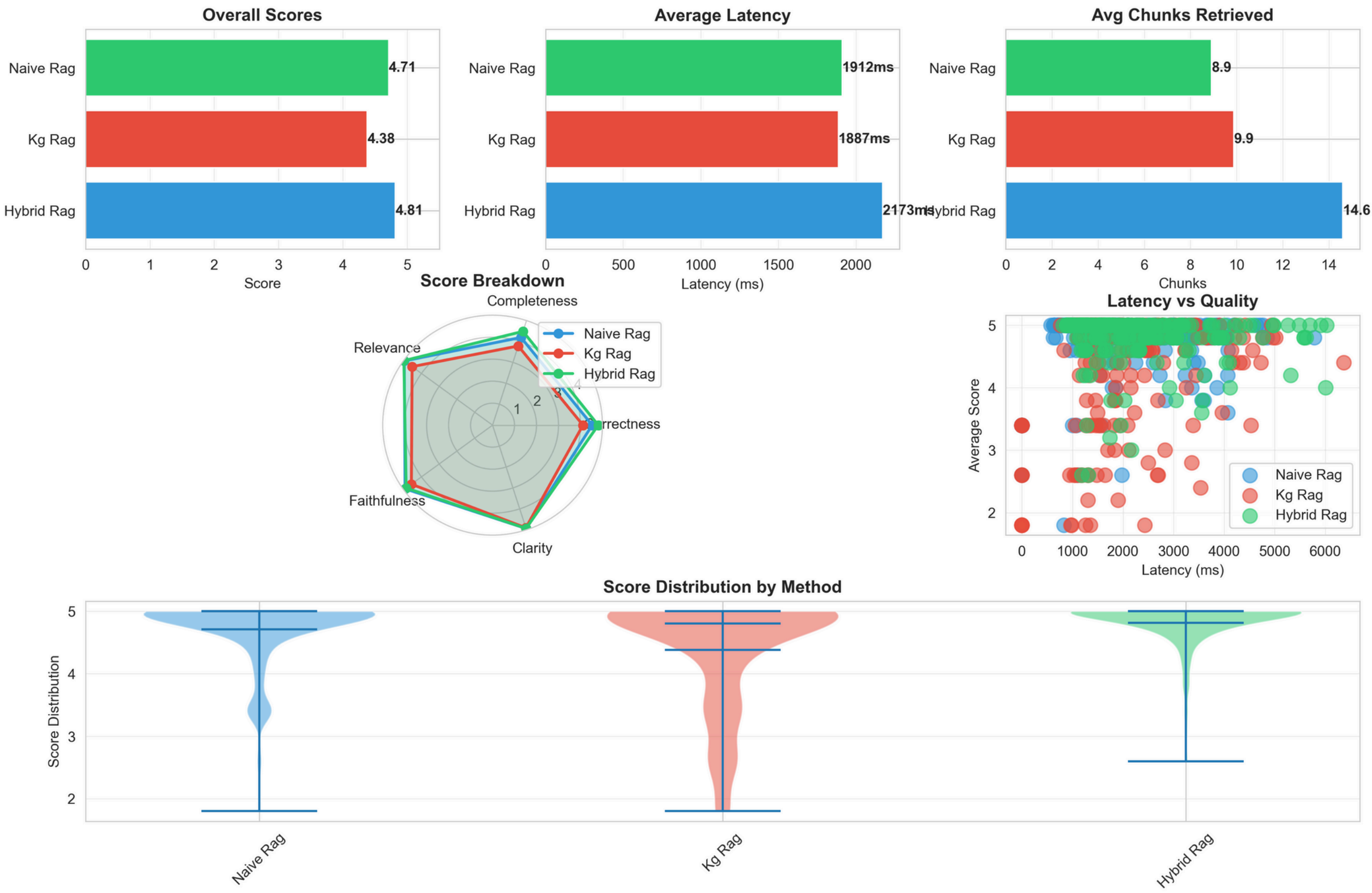
	Method	Overall Score	Correctness	Completeness	Relevance	Faithfulness	Clarity	Avg Latency (ms)
1	Hybrid Rag	4.81	4.807843	4.482353	4.988235	4.858824	4.937255	2172.54
2	Naive Rag	4.71	4.533333	4.192157	4.964706	4.921569	4.941176	1911.85
3	Kg Rag	4.38	4.113725	3.784314	4.525490	4.564706	4.905882	1886.91

DETAILED STATISTICS

	method	latency_ms_mean	latency_ms_std	latency_ms_min	latency_ms_max	num_chunks_mean	num_chunks_std	num_chunks_min
Hybrid	Rag	2172.54	1169.38	820.14	6028.12	14.60	5.94	1
	Kg Rag	1886.91	1066.34	0.00	6363.28	9.88	7.10	0
Naive	Rag	1911.85	1143.08	0.00	5780.17	8.92	6.91	0

Qualitative Results

Comprehensive RAG Method Comparison Dashboard



Demo!